Improved Combined Scatterers Interferometry With Optimized Point Scatterers (ICOPS) for Interferometric Synthetic Aperture Radar (InSAR) Time-Series Analysis

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***Abstract*— Despite the development of point-based interferometric synthetic aperture radar (InSAR) time-series methods, such as persistent scatterer interferometry (PSI), it remains difficult to identify reliable measurement points (MPs) in areas under vegetated cover. In this study, we developed a new algorithm, improved combined scatterers interferometry with optimized point scatterers (ICOPS), to monitor changes in surface deformation by combining persistent scatterers (PSs) and distributed scatterers (DSs). The algorithm was subsequently improved through the application of a machine learning process. A generalized likelihood ratio test (GLRT) was applied to identify statistically homogeneous pixels suitable for a small SAR dataset. A machine learning-based optimization process was used to overcome the increase in MPs after the combined scatterers interferometry (CSI) process. The MP optimization process used the support vector regression (SVR) algorithm to find the optimal point from all MPs in the dataset. To improve the reliability of the MP surface deformation mapping after optimization, the optimized hot-spot analysis (OHSA) method was applied to obtain spatially clustered MPs; this was possible despite a low deformation identification rate. To demonstrate the effectiveness of the ICOPS method, we applied this approach to Yellowstone Lake in the USA using 95 SAR images from the Sentinel-1 satellite that was taken during the period of 2017–2020. The CSI method produced an increase in MP density, especially in areas not covered by PSI, which indicated its ability to detect deformations in mountainous areas. Comparisons with GPS data and traditional methods produced promising results with an accuracy of 1 cm/year in terms of the root mean square error (RMSE). The optimization process in CSI also had the advantage of retrieving helpful information in the MP dataset and increased the accuracy of CSI by 12%. Deformation mapping using the optimized results provided new insights regarding the spatial clustering of surface deformation in MPs. This could provide the foundation for developing a posttime-series optimization process based on point scatter as a surface change detection tool.**

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I. Introduction

OVER the past few decades, the measurement and detection of ground deformation have been achieved using interferometric synthetic aperture radar (InSAR), which provides broad coverage, high resolutions, and frequent acquisitions [1], [2]. In its early application, small ground deformations could be measured using differential InSAR (DInSAR) [3], [4]. However, DInSAR is limited by temporal decorrelation and atmospheric disturbance, which affects the estimation of ground displacement. DInSAR cannot produce a long-term deformation time series; it can only provide its acquisition pair date. Thus, for the past two decades, InSAR time-series techniques have been developed to obtain better measurements and reduce the effect of this limitation. In general, InSAR time-series techniques can be grouped into two categories, persistent scatterers (PSs) and distributed scatterers (DSs), according to the scattering mechanisms of the ground target [5].

Generally, the PS method focuses on point-like coherent targets that are controlled by single or multiple scattererers. These targets exhibit highly stable backscattering behavior, usually corresponding to artificial reflectors, man-made structures, or bare rocks [6]. The PS method adopts a single reference image to generate interferograms from single-look complex images that are coregistered to a common reference image. This includes several widely used algorithms, such as PSInSAR [7], StaMPS [8], [9], and IPTA [10]. In contrast to the PS method, the DS method uses distributed targets that contain many small random scatterers, and none of them are dominant [11]. Distribution targets are primarily found in rural environments, such as bare land, sparse vegetation, or mountainous areas. In the DS method, multireference interferograms with short time-space baselines are generated. This classical method is referred to as the small baseline subset (SBAS), which is widely applied in rural areas where the PS

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method cannot be applied for robust analysis [12], [13]. A new generation of InSAR time-series technology was developed to maximize the spatial sampling of deformation signals and overcome the limitations of point-scatterer methods. One method that has been proposed to overcome these problems is SqueeSAR,[1] developed by Ferretti *et al.* [14]. Unlike the common time-series algorithm, SqueeSAR processes PS and DS targets to jointly retrieve the deformed signal on the basis of their statistical properties. In particular, this method makes full use of all possible combinations to preprocess the DS target without restrictions according to time or geometrical baselines.

Although it results in an increased density of measurement points (MPs), the combination of PS and DS targets may also reduce the efficiency of analysis, especially over a broad study area. Therefore, it is challenging to determine how to extract useful information from a large dataset. To determine which points are most reliable, Lu *et al.* [15] proposed a method for the analysis of persistent scatter interferometric hot spots and cluster analysis. This was based on the Getis-Ord Gi* statistical data spatial clustering method, which is used to collect PS hot spots [15]. However, manual intervention remains necessary in this method, which can reduce the efficiency of the analysis, especially for large PS/DS MP datasets. In addition, a new approach of optimized hot-spot analysis (OHSA) has been developed for PS and DS; its potential was evaluated in the detection of landslides in the Volterra area in the central Tuscany region of Italy [16]. Compared with persistent scatter interferometric hot spots and cluster analysis, OHSA has a significant advantage in that the scale distance used for Getis-Ord Gi* statistics can be automatically determined by the incremental spatial autocorrelation analysis.

Machine learning has been widely used to detect deformations for various purposes in the InSAR time-series process. For example, Brengman *et al.* [17] used a deep learning algorithm to automatically detect deformations in volcanoes [15]; Li *et al.* [18] used the support vector regression (SVR) algorithm to predict surface subsidence in a mining area based on a deformation time series. The application of machine learning algorithms to deformation time-series datasets to evaluate and obtain optimal MPs shows promise results. The SVR algorithm as optimization machine learning allows improving point scatterers reliability by separating points based on their characteristics. Based on hyperplane assumptions, the SVR algorithm can be more flexible in finding solutions and suitable for complex datasets.

In this study, a new approach to measuring deformation was proposed: improved combined scatterers interferometry with optimized point scatterers (ICOPS). The proposed method was constructed based on combining PS and DS points to perform CSI. We then optimized the point scatterer to increase the reliability of measurements based on a machine learning-spatial approach. The ICOPS method was applied in a surface deformation study at the Yellowstone National Park using Sentinel-1 SAR datasets for the period of 2017–2020. First, a comparison between the CSI and SBAS deformation time-series results was conducted to identify the advantages offered by CSI. The results of deformation measurements using the CSI method were then validated using *in situ* GPS measurements. Furthermore, a comparison of the CSI deformation measurement with the ICOPS method was conducted to evaluate the optimization approach with the SVR in the time-series process. Finally, we conducted an analysis of the spatial changes and possible factors affecting deformation around the study area.

[1]Trademarked.

II. Data and Study Area

A. Study Area

The Yellowstone Plateau volcanic field forms a high continental gap between the central and northern Rocky Mountains. The plateau has a mean elevation of approximately 2400 m and is surrounded by mountains; mountainous plains surround the southwest, reaching elevations of 3000–4000 m. According to geographic analysis, two structural revival domes were formed after the last eruption to produce the crater: the Mallard resurgent dome in the southwestern area of the Yellowstone Crater and the Sour Creek resurgent dome in the northeast [19]. Both tectonic and hydrothermal processes drive frequent earthquakes that can lead to surface deformation at different rates. Before the eruption in the Yellowstone Plateau, the surrounding mountains and valleys were generally in their current structural form with minor fluctuations in deformation. The density flow of pyroclasts produced during the eruption formed the crater, which spread outward, such that it filled the valley and deposited igneous tuff in the surrounding mountains.

Although there have been few magma eruptions around the Yellowstone Caldera in the past 70 years, GPS and InSAR have been used to detect volcanic activity and surface deformation changes [20]. Later, GPS measurements showed that the crater entered a sinking stage until 2005 when the crater returned to the extreme uplift stage. The largest vertical movement was recorded at the White Lake GPS station on the eastern edge of the crater. From 2004 to 2010, the total uplift was approximately 27 cm. The inflation rate slowed in 2008, and the crater began to recede again in the first half of 2010 [21].

The ICOPS method was applied to assess a time series of changes in surface deformation in the Yellowstone National Park with Sentinel-1 SAR datasets. The Yellowstone National Park covers parts of Wyoming (WY), Montana (MT), and Idaho (ID); it is highlighted as the site of volcanic activity in Fig. 1(a) [22]. This study focused on the northern part of the park near the Sour Creek resurgent dome, which is shown in Fig. 1(b). To support the method and validation of the results generated by the study, GPS data from the LKWY station were used in this study area, as shown in Fig. 1(b) [23].

B. Synthetic Aperture Radar Data Processing

We collected 95 SAR images that had been obtained from the Sentinel-1 *C*-band satellite to detect surface deformation in Yellowstone for the period of 2017–2020. The SAR images

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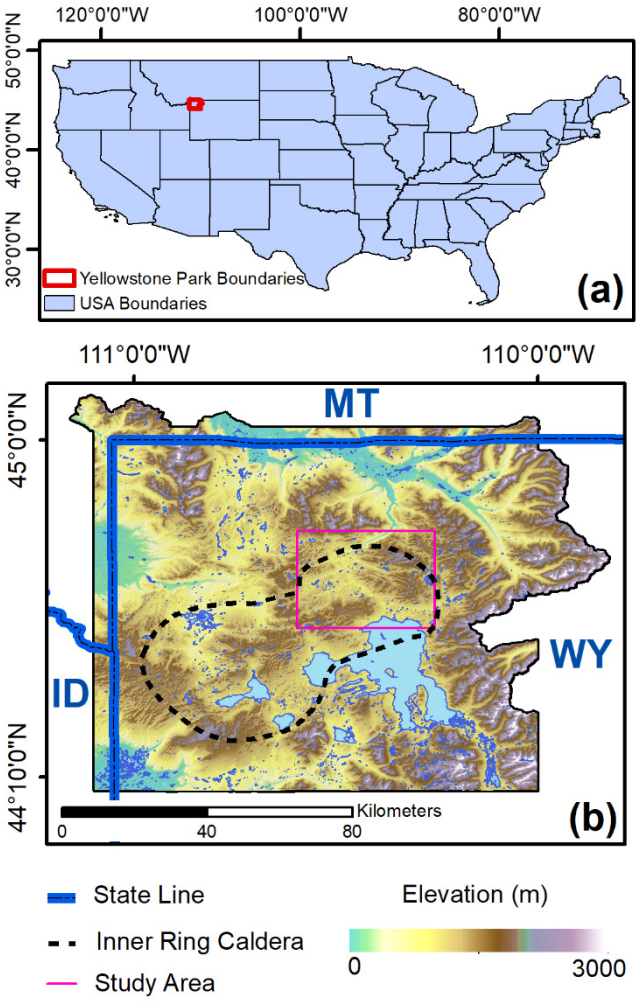


Fig. 1. (a) Location of the Yellowstone National Park marked in the red box, with state boundaries marked by blue lines. (b) Image of the Yellowstone National Park produced with a digital elevation model from the Shuttle Radar Topographic Map (SRTM). The Black shaded line shows the boundary of the inner caldera [24], and the pink rectangle represents the study area.

obtained from ascending tracks were 95 images taken from path number 49. The Sentinel-1 satellite has an observation repeat cycle of 12 or six days. This short temporal interval has the advantage of maintaining coherence despite short radar wavelengths (approximately 5.5 cm) [25].

In a multitemporal InSAR analysis, a subset of pixels (PS or DS) is selected based on particular statistical data (e.g., amplitude or phase on multiple SAR images); the mean deformation rate of these selected pixels is estimated [6]. The main steps in the new method included the generation of interferograms, selection of PS and DS, 3-D phase expansion, and calculation of the mean deformation velocity and displacement time series. We will detail these main steps in this section. We generated an SAR interferogram with a small temporal and perpendicular baseline. Compared with SqueeSAR [14], which makes full use of all possible interferograms, this reduces the interferogram number and processing time.

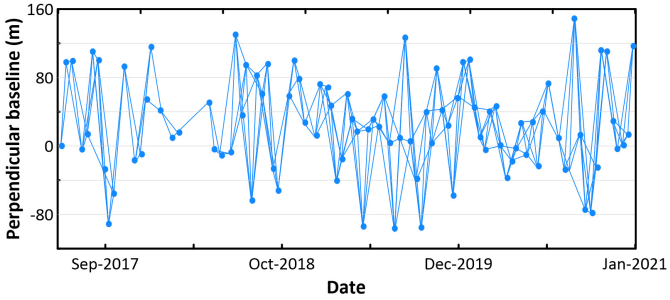


Fig. 2. Perpendicular baseline of pairs formed in the Sentinel-1 SAR dataset during the interferogram process. Blue circles represent the acquisition dates of collected SAR images.

Using Sentinel-1 data, we selected pairs of images with a maximum time interval of 60 days and a perpendicular baseline of 160 m. Fig. 2 shows the perpendicular baseline used in this study, which was generated from the Sentinel-1 datasets. Before interferogram generation, the SAR data underwent a coregistration process, in which two SAR images were aligned to subpixel accuracy for precise determination and noise reduction; these images formed an interferometric pair. Then, SAR images were resampled to match secondary images with reference images. After the coregistration process had been completed and before the interferogram generation process had been initiated, coregistered images were cropped to focus on the study area. A topographic phase was generated in the interferogram processing step. After the interferogram image had been generated, the Shuttle Radar Topographic Map digital elevation 1-arcsecond model (SRTM DEM 1 arcsec) was used as a reference to subtract the topographic phase from the interferogram [26]. After removal of the topographic phase, a DInSAR phase was generated, containing the deformation phase and atmospheric artifacts.

III. Method

This study implemented the ICOPS method in an InSAR time-series analysis of the SAR dataset. We then applied the SVR algorithm to evaluate CSI MPs for the acquisition of optimal MPs. After optimal MPs had been obtained, an OHSA was applied to improve the reliability of mapping with spatial clusters. The general workflow of the ICOPS method developed in this study is shown in Fig. 3. PS targets were obtained using the StaMPS procedure, while DS targets were obtained using a modified SqueeSAR-based approach. As an alternative to the Kolmogorov–Smirnov test or the Anderson–Darling test, the generalized likelihood ratio test (GLRT) test was applied to identify statistically homogenous pixels (SHPs). That method was recently applied and has concerned the sensitivity on few datasets and large time processing. However, the GLR test provides flexibility and improvement as a testing method for SHP. The use of the GLRT enabled the SHP identification process to proceed with fewer images while preserving the ability to acquire good results [27]. The identified DS points were MPs complementary to the PS points in the analysis of surface deformation changes. Then, DS and PS were combined for phase unwrapping. The singular

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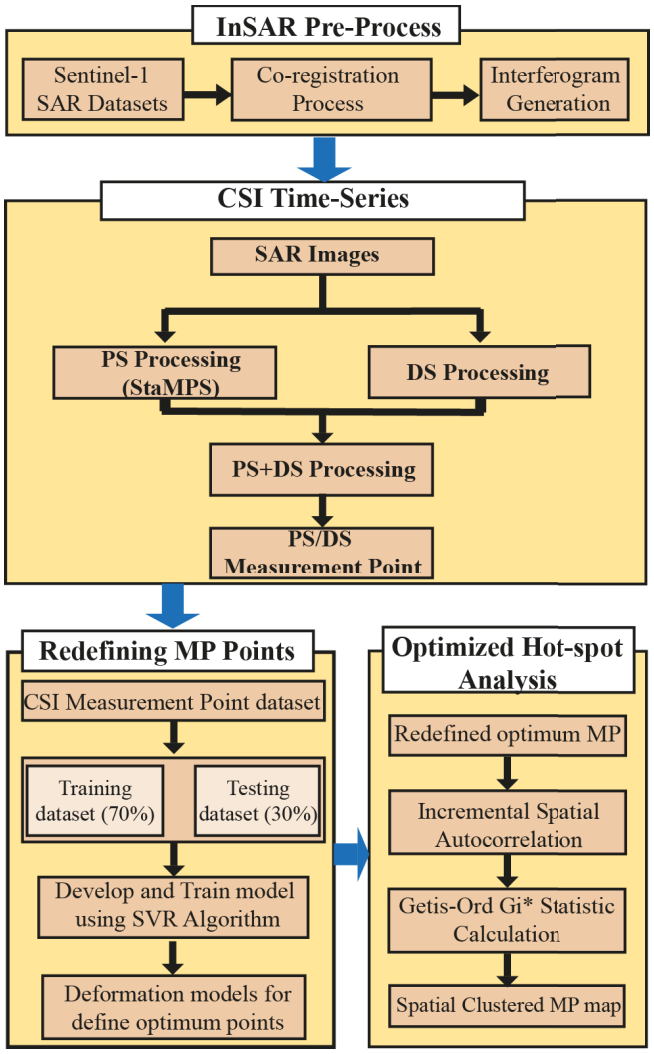


Fig. 3. Workflow of ICOPS method developed in this study.

value decomposition approach was applied to the deformation estimation to obtain a deformation time-series map [28]. With the increasing number of MPs obtained from the CSI process, we implemented an optimization process to obtain better MPs. The SVR machine learning algorithm was used to optimize MPs by evaluating deformation time-series information. Then, the OHSA technique was applied to the optimized data, based on the Getis-Ord Gi* statistic; this improved the detection of slow surface deformation rates via spatial clustering of MPs. The proposed optimization process overcame limitations in the analysis of the deformation map. The steps in the ICOPS algorithm are explained in greater detail in the following.

A. Combined Scatterers Interferometry

In this study, a new algorithm for an InSAR time-series analysis was proposed to measure surface deformation by combining PS and DS in SAR data. There have been many studies of the combination of PS and DS in SAR data [16], [27], [29], [30], but few algorithms of InSAR data deformation measurements have been combined with a machine learning algorithm. Therefore, in this study,

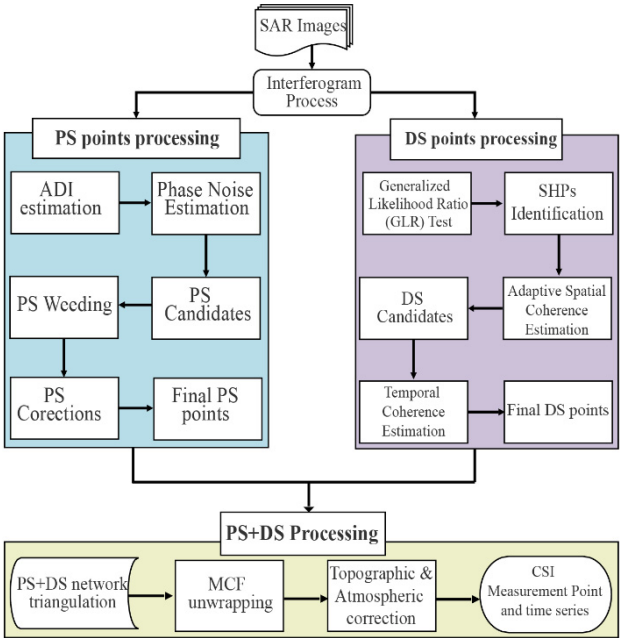


Fig. 4. CSI flowchart modified from Wang *et al.* [29].

we combined the CSI method with machine learning to optimize the CSI result. The CSI consisted of three steps (as shown in Fig. 4): PS processing, DS processing, and a joint analysis of PS and DS. The PS and DS processes focused on the selection of PS and DS targets, as well as estimation of the optimal phases. The selected PS and DS points were then processed together to generate the deformation measurements. This step is discussed in detail in the following.

1) Persistent Scatterers Processing: The first step, PS processing, is the same procedure as the approach used in the StaMPS algorithm to select PS points [31]. The StaMPS algorithm was selected as the base algorithm in the CSI because of its advantages in terms of surface deformation detection in natural terrain. The StaMPS algorithm assumes that the deformation and interferometric phases are spatially correlated; therefore, this algorithm does not rely on the assumption of ground deformation caused by a temporal factor. This advantage has resulted in the generation of more PS points in rural or natural areas compared with the use of other PS algorithms. The selection process of the StaMPS algorithm is based on phase characteristics and is suitable for locating low amplitude natural targets with phase stability, as shown in the following equation [31]–[33]:

$$D_A = \frac{\sigma_A}{\mu_A} \tag{1}$$

where D_A indicates the amplitude dispersion value, while σ_A and μ_A are the standard deviation and mean of the amplitude values, respectively. In the StaMPS algorithm, points with $D_A < 0.4$ are selected as PS candidates [8], [29]. The StaMPS algorithm then reselects the PS candidates by employing an iterative model in which the coherence of the PS candidates is calculated using other nearby PS candidates; points with low coherence values are rejected [6].

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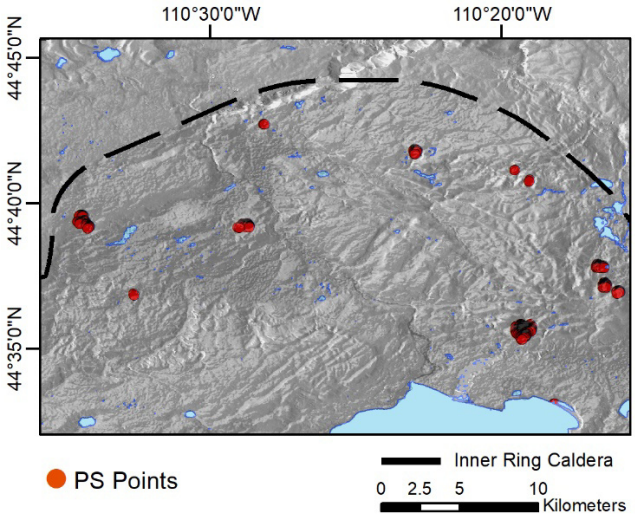


Fig. 5. Point scatterers generated from the StaMPS algorithm with SAR amplitude image layover.

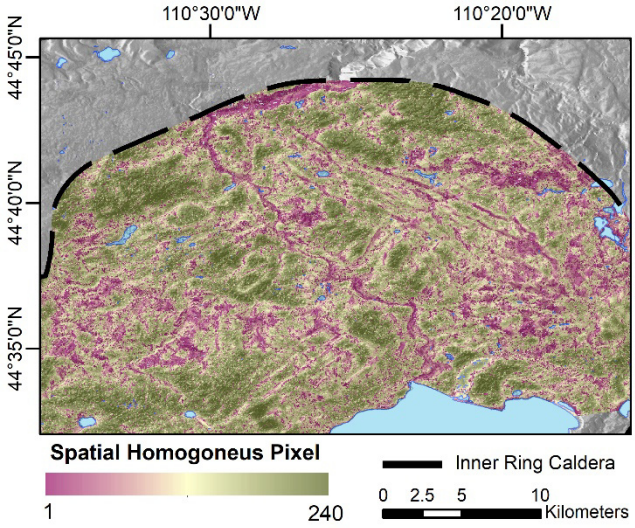


Fig. 6. SHPs of the study area generated by a GLRT.

After the PS had been selected, PS weeding is proceeded by removing the PS that was in close proximity to each other, resulting in deformation smoothing over time. The spatially uncorrelated digital elevation model error in the remaining PS was corrected. In total, 4042 PS points were generated in the study area, as shown in Fig. 5. PS points generated using the StaMPS algorithm were mostly found in areas near Sulphur Mountain within the lava creek tuff bedrock layer from past eruptions in Yellowstone. The area mainly contains barren land with some herbaceous cover (resulting in a high amplitude dispersion value), while the other parts of the study area were primarily covered in evergreen forest and shrub (resulting in a low amplitude dispersion value); the parts covered in evergreen forest and shrub were discarded from the PS selection. The PS that had already been selected was jointly processed with DS points from the second step before advancement to the third step of unwrapping and deformation time-series estimation.

2) Distributed Scatterers Processing: The selection of DS points began with the identification of SHPs for which amplitudes exhibited similar statistical characteristics over time [14], [29]. For an arbitrary pixel, a goodness-of-fit test is usually employed to identify the surrounding SHPs within a specific window [34]. In this study, SHP testing was performed using a GLRT in the fast statistically homogeneous pixel selection (FaSHPS) algorithm [34]. The GLRT is a test statistic determined by the ratio of the pdfs of pooled datasets evaluated at the maximum likelihood estimate of parameters established in the alternative and null hypotheses. The GLRT was used in this study because it had better detection rates in different simulation scenarios, compared with other test statistics [35], [36]. For an estimate of the Rayleigh parameter, we used the maximum likelihood estimate of the dataset parameters, as expressed in the following equation:

$$\hat{\sigma}^2 = \frac{1}{M}\sum_{i=1}^{M} x_i^2, \quad x_i = \text{sample data} \tag{2}$$

with the hypothesis assumption described in the following equations:

$$H_0 : \hat{\sigma}^2 = \hat{\sigma}_p^2 = \hat{\sigma}_q^2 \tag{3}$$

$$H_1 : \hat{\sigma}_p^2 \neq \hat{\sigma}_q^2. \tag{4}$$

The identification of SHPs was based on the similarity of the amplitude values over time; thus, in (3), H_0 was the correct hypothesis assumption. Then, the logarithm for the GLRT statistic under the similar value of both points was expressed according to the following equations [35]:

$$T = 2M\ln\left(\hat{\sigma}^2\right) - M\ln\left(\hat{\sigma}_p^2\right) - M\ln\left(\hat{\sigma}_q^2\right) \tag{5}$$

$$\hat{\sigma}^2 = \frac{\hat{\sigma}_p^2 + \hat{\sigma}_q^2}{2}. \tag{6}$$

Fig. 6 shows the results of SHP calculations conducted using the GLRT in the study area. The processes were applied in a window size of 15 × 15 pixels. With the aim of getting a reliable pixel estimate, the use of a window size of 15 × 15 has been fulfilled in our case. The use of lower or too high window size can lead to unreliable or overestimation of coherence that does not provide a significant change to the result. The process began by selecting and analyzing a single pixel; subsequently, the other pixels were processed [37]. Then, those pixels with a similar test result became "brothers," while adjacent pixels with different test results were discarded. Next, adjacent pixels were analyzed to determine whether there were similar signal characteristics over the entire study area, based on the SAR data. The signals were averaged and applied to form a DS candidate based on the SHP value. SHPs with values >20 were selected for subsequent processing [14].

Then, the adaptive spatial coherence of the possible interferogram was estimated. Fig. 7 shows the resulting adaptive coherence map, determined using the process expressed in the following equation:

$$\mu_0 = \frac{\left\langle x_1 x_{2\cdot\text{SHP}}^* \right\rangle}{N\sqrt{\left\langle |x_{1\cdot\text{SHP}}|^2 \right\rangle \left\langle |x_{2\cdot\text{SHP}}|^2 \right\rangle}} \tag{7}$$

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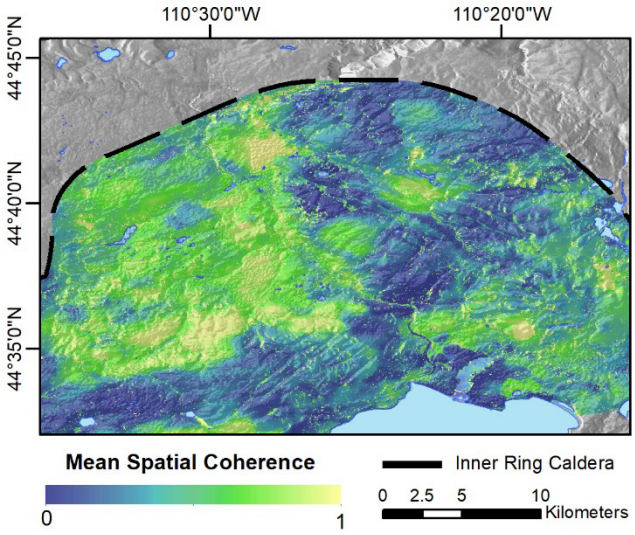


Fig. 7. Adaptive spatial coherence map of the study area.

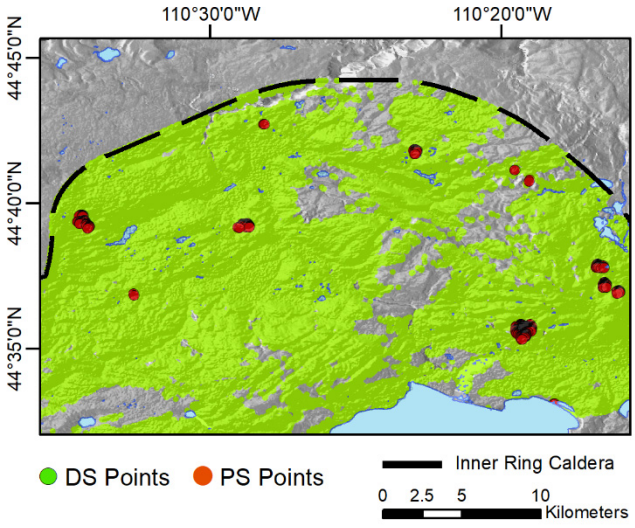


Fig. 8. PS and DS points acquired in the study area using the combined process in the CSI analysis.

where μ_0 is the adaptive spatial coherence, N is the number of data points, x_1 and x_2 are two complex SAR pixels of both acquisition dates, and $x_1 x_2^*$ is the complex conjugate value of x_1 and x_2. Points with spatial coherence $\mu_0 > 0.2$ were selected as DS candidates to increase pixel density in determining candidate DSs.

After DS candidates had been selected based on SHPs and the estimation of adaptive spatial coherence, temporal coherence was estimated. This process was conducted to select pixels with a stable phase over time. The process is expressed according to the following equations [37]:

$$\gamma_T = \frac{1}{M}\left|\sum_{i=1}^{M} \exp\left[j\left(\Delta\phi_{\text{ori}} - \Delta\phi_{\text{rec}}\right)\right]\right| \tag{8}$$

$$\Delta\phi_{\text{ori}} = \Delta\phi_{\text{def}} + \Delta\phi_{\text{topo}} + \Delta\phi_{\text{orbit}} + \Delta\phi_n \tag{9}$$

$$\Delta\phi_{\text{rec}} = \frac{4\pi}{\lambda}\left(\Delta d - \bar{\Delta d}\right) + \frac{4\pi}{\lambda}\frac{B_\perp}{r\sin\theta}\left(\Delta z - \bar{\Delta z}\right) + \left(\Delta\phi_n - \bar{\Delta\phi_n}\right) \tag{10}$$

where γ_T is the temporal coherence estimation; M is the amount of data processed; and $\Delta\phi_{\text{ori}}$ and $\Delta\phi_{\text{rec}}$ are the original interferometric phase and reconstructed interferometric phase, respectively. The difference between the original and reconstructed phases represents the noise level of the SAR phase history $\Delta\phi_n$; it is known as the temporal coherence value.

In total, 1141 316 DS points were selected on this DS process. We selected DS candidate points with $\gamma_T > 0.75$; this threshold was selected to show a better quality of pixel characteristics with a value close to 1. For further CSI analysis, together with the PS points, these DS candidate points were used for unwrapping and deformation time-series estimation. During this process, the area not detected by PS was eventually detected by DS. The undetected DS area was located close to Thistle Creek and Sulphur Mountain, where most of the evergreen forest was concentrated.

3) Combined PS-DS Process: As shown in Fig. 8, compared with the PS points, abundant DS points were detected, indicating that the selection of MPs was substantially improved by the inclusion of DS points in this study. A DS point with an estimated optimal phase value was equivalent to a quasi-PS point; thus, it could be combined with its corresponding PS point for further analysis. DS points that coincided with PS points were removed. After completion of PS and DS point selection, as well as the combined PS-DS process, phase stacks on selected pixels were obtained from both algorithms.

Adaptive filtering was applied to each interferogram dataset to enhance the signal-to-noise ratio of the interferometric phase of the interferogram. This adaptive filtering was used instead of rectangular averaging to normalize a particular pixel to its neighboring SHPs [29], [34]. This algorithm proved to be more efficient without the requirement to use all possible interferograms, as described in the study by Ferretti *et al.* [14], which used the SqueeSAR algorithm. After adaptive filtering had been implemented, the selected phase stack pixel data were unwrapped using a minimum cost flow algorithm. The minimum cost flow approach connects and discharges residues by connecting them with branch cuts to minimize the total weighted length [38]. The unwrapped results were then used in the final process of retrieving mean deformation rates and a deformation time series. This process was carried out using the singular value decomposition approach to derive a deformation time-series measurement based on a minimum norm criterion of the deformation rate [12], [39]. The process was also refined by correction of phase unwrapping error when phase jumps were detected, correction of atmospheric artifacts and topographic errors through an iteration process, and mitigation of temporal noise using a finite-difference smoothing approach.

B. SVR Algorithm

The SVR is a machine learning algorithm that was proposed by Vapnik [41]. It estimates the connection between the input and the output of a system from the existing sample [10].

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This method examines correlations by transferring data to a higher dimension using (11) and then solves the equation with structural risk minimization based on (12). When the relationship is estimated accurately, it can be used to predict the outputs from the inputs [11]

$$f(x) = w\varphi(x) + b$$

$$\min : \left[\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i + \xi_i*\right] \tag{11}$$

$$\text{s.t.} \begin{cases} y_i - (w\varphi(x_i) + b_i) \le \varepsilon + \xi_i \\ (w\varphi(x_i) + b_i) - y_i \le \varepsilon + \xi_i* \\ \xi_i, \quad \xi_i* \ge 0. \end{cases} \tag{12}$$

In the training dataset (x_i, y_i), $i = 1, 2, 3, \ldots, n$, where x_i is the input vector, y_i shows the corresponding output, and i is the ith training sample. In addition, n is the data size, and $f(x)$ is applied to calculate the output based on the input data. Furthermore, $\varphi(x)$ is an irregular function for mapping input data to high-dimensional feature space.

In this study, the parameters of the SVR algorithm (gamma, epsilon, and C) were determined using the gray wolf optimization algorithm, which can evaluate training and testing datasets in the form of deformations at each point in the CSI results [40], [42]. The optimization algorithm was used to reduce limitations in the manual determination of SVR parameters. In the preparation stage, the inventory dataset was determined by calculating the correlation and linear coefficients for each deformation MP derived from the CSI result. Then, an MP with a linear coefficient value above the median value and correlation coefficient >0.7 was marked as "1." An MP with a linear coefficient value below the median value and a correlation coefficient <0.7 was marked as "0." For the prediction model training process, the dataset with attributes of "0" and "1" was randomly divided into a training dataset (70%) and a testing dataset (30%) [43]. Then, the machine learning training process was carried out using the SVR algorithm to generate model deformation. The optimization process is carried out based on the distribution of the modeling results from the SVR algorithm. This distribution is used to identify areas with an optimum point (indicating deformation) or nonoptimal deformation point. Then, the results of the evaluation of the correlation deformation time series at each point are taken into consideration in the optimization process. The SVR machine learning method was used to optimize the MPs that were obtained in the time-series analysis process and reduce the number of insignificant MPs that could interfere with the deformation results. The development of this method represented a preliminary advance in the postprocess optimization of time-series methods in general.

C. OHSA Method

The OHSA method was applied to the optimal PS-DS points from datasets of Sentinel-1 images. This process was conducted on PS-DS MPs that had been derived from the optimization using SVR to enhance the spatial availability of MPs based on the Getis-Ord Gi* statistics calculation. Fig. 9 shows the general workflow of the OSHA method [16]. The

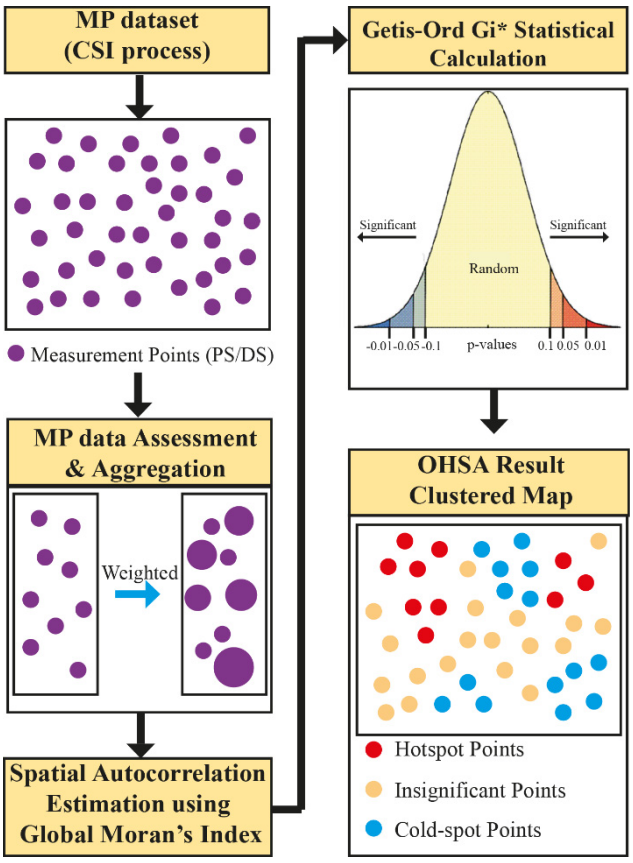


Fig. 9. Workflow of the OSHA method for spatial clustering of MPs modified from Lu *et al.* [16].

method required an adequate number and variation of MPs for the analysis to proceed. The first step was the assessment and aggregation of PS and DS data, which ensured that the OHSA had an adequate number and variation of data points in the analyzed dataset. The deformation rate was used as a medium for the analysis of PS/DS MPs. We then calculated the mean nearest neighbor from each PS/DS MP and evaluated the distribution of all mean nearest neighbor distances. We also evaluated the median nearest neighbor for each PS/DS MP. Based on the mean nearest neighbor and median nearest neighbor evaluation, each PS/DS MP was converted into a velocity-weighted data point for the OHSA analysis process [16].

The method was able to identify an appropriate scale of analysis to determine the Getis-Ord Gi* statistic. A notable challenge in traditional hot spot analysis involves the determination of the scale distance (d) for the Getis-Ord Gi* statistic [15]. To address this challenge, an incremental spatial autocorrelation analysis was applied in this study. An effective approach to determining scale distances is consideration of the statistical nature of the dataset based on the distance at which a high spatial autocorrelation is found [44]. This approach can usually be accomplished using an iterative process; data-driven characteristics determine how the spatial autocorrelation varies according to distance. The spatial autocorrelation was calculated using the global Moran's index, as shown in the

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following equation [45]:

$$I = \frac{n}{S_0} \frac{\sum_{i=1}^{n} \sum_{i=j}^{n} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_{j=1}^{n} (x_j - \bar{x})^2} \tag{13}$$

where n is the total number of MPs, $\bar{x}$ is the mean velocity of a derived MP, w_{ij} is the spatial weight, and S_0 is the sum of all spatial weights derived from the following equation:

$$S_0 = \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}. \tag{14}$$

The outcome of the global Moran's I index calculation was used to assess the significance of the spatial autocorrelation by calculating the z-score using the following equation:

$$z = \frac{I - E[I]}{\sqrt{\text{Var}[I]}}. \tag{15}$$

The spatial autocorrelation was estimated at each scale distance from (15); this value was used to calculate the Getis-Ord statistic for MPs. The single PS point at location i and its neighbors within a searching distance d for every single MP at site I (i.e., the Gi index) was calculated using the following equation [46]:

$$G_i^*(d) = \frac{\sum x_j + x_i + n_{ij} \times \bar{x}^*}{\sqrt{s^* \left\{ \left[\left(n \times n_{ij} \right) - n_{ij}^2 \right] / (n-1) \right\}}} \tag{16}$$

where n is the total number of PS; n_{ij} is the number of PS within the scale distance d; x is the surface displacement recorded PS/DS; and x^* and s^* define the mean and standard deviation of all displacements revealed by the whole PS, respectively. Random clusters and patterns could be identified using these Getis-Ord Gi* statistic values. The Getis-Ord Gi* statistic for each PS/DS MP was the calculated value of the z-score (standard deviation) and p-value (independent probability), both of which were used to estimate the statistical significance of spatial autocorrelation. Specifically, each MP could be clustered based on the confidence level and z-score for cluster statistics at 90%, 95%, and 99%, with p-values of 0.10, 0.05, and 0.01, respectively. In determining the final optimization process using hotspot analysis on the surface deformation detection cluster, we use the z-score and the p-value. The MP with a value of less than 0.5 cm/year is ignored in the optimization process because it exceeds the standard deviation of 0.5 cm. Then, the OHSA process shows the results with the MP configuration that has a deformation rate above 0.5 cm/year, p-value of 0.01, and z-score >2.71. The statistical calculations of the positive and negative z-score showed that the MP spatial cluster moved toward and away from the line of sight, respectively, in terms of SAR applications.

The next step involved increasing the detection reliability of the mean deformation rate for each time-series method on the distribution map. The OHSA method evaluated the mean deformation rate data for the MPs and then clustered these data. This improved the analysis of the deformation data contained in the clustered MPs. The results of this spatial clustering also improved deformation detection, especially in areas with low levels of deformation, such as landslides and areas with infrastructure damage. The resulting MP information could be used to effectively manage the deformation time-series measurement data.

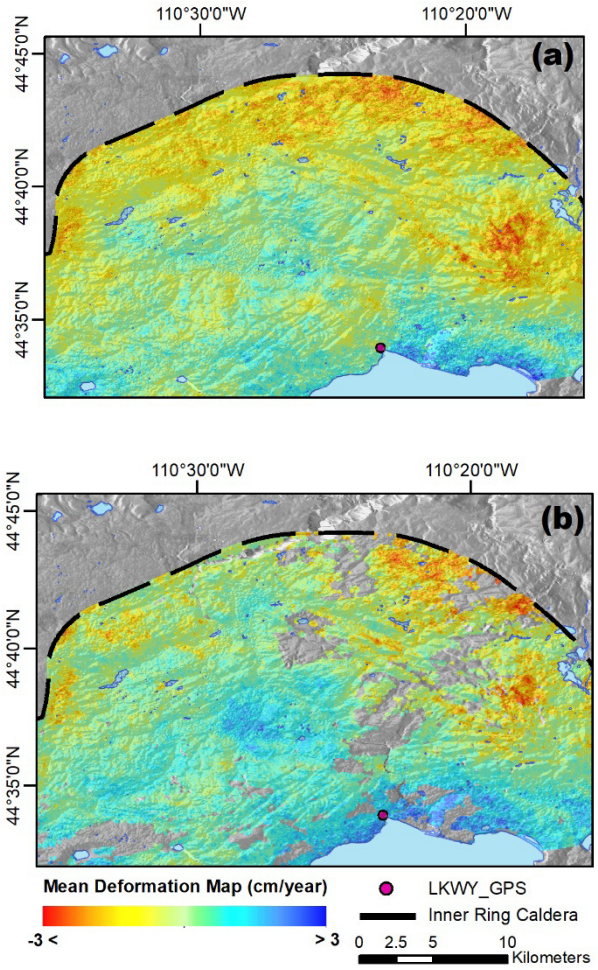


Fig. 10. Mean deformation map from Sentinel-1 SAR data for 2017–2020: (a) SBAS and (b) CSI.

IV. Results

A. Comparison of Mean Deformation Maps Produced Using Different Methods

Surface deformation was measured using different InSAR time-series analysis methods in the Yellowstone National Park. The deformation signal obtained by CSI was clearer than the deformation signals estimated by the PS interferometry (PSI) and SBAS methods. As expected, the proposed CSI method recognized more MPs than StaMPS. Unlike SBAS, the CSI method could recognize MPs regardless of the terrain characteristics; this resulted from the MP analysis with SHP. The PS number detected by the ascending Sentinel-1 image was 4042, while the DS number was 1 141 316. The DS number was significantly larger than the PS number, which suggests that the number of MPs could be significantly increased by including DS.

Fig. 10 shows the mean vertical deformation maps produced by SBAS [see Fig. 10(a)] and CSI [see Fig. 10(b)] methods

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using the SAR dataset from the Sentinel-1 satellite. Based on the mean deformation map, the PS points were distributed in areas that had a stable phase, while SBAS indicated that all points were within the study area. The use of the proposed CSI method revealed the MP distribution based on a combination of PS and DS, which effectively described the deformation distribution. The CSI method could detect the deformation distributions in stable phase conditions (hard rocks) and in areas with land that lacked dense vegetation cover. We compared the distributions of CSI points with land cover data to determine the MP distributions on these land types. Based on landcover data provided by the United States Geological Survey in 2011, 35% and 26% of CSI points were in forest and meadows, respectively [47], [48]. These results indicated the usefulness of the CSI method for mapping deformations in mountainous areas and under vegetation cover, based on the distribution of DS incorporated in the analysis process.

B. Machine Learning Optimization for Time-Series Analysis and Validation

A deformation analysis optimization process using machine learning (SVR) and the OHSA method was conducted in the study area. The CSI analysis detected 1 145 356 MPs in the area using Sentinel-1 data for 2017–2020. Many of the MPs were found on bare land and exposed bedrock [12]. After the optimization process had been conducted using the ICOPS method, the number of optimal MPs decreased to 735 610; there were many insignificant MPs subtracted during the optimization process, which are shown in Fig. 11(a). Fig. 11(b) shows the MP optimization results that were obtained using ICOPS. The SVR algorithm performed calculations based on the correlation and linear coefficients of the time-series data for each MP. This had an impact on the optimized deformation time-series graph analysis process. Based on the MP results for the OHSA method, in which the SVR algorithm had optimized, the deformed areas could be effectively clustered. According to the calculation of the Getis-Ord Gi* statistic, the mean deformation values were spatially clustered based on z- and p-values. Therefore, the deformation area (especially the small deformation range) could be effectively identified after this optimization process. Areas that had experienced subsidence were identified around the Sour Creek hills, which may have been influenced by groundwater conditions and hydrothermal activity [21]. The uplift deformation detected at the edge of Yellowstone Lake may have been influenced by fluid migration underground and sediment consolidation, especially near the Indian Pond and Turbid Lake areas. The deformation signal was spread over areas with land that lacked substantial vegetation cover, which matched the characteristics of the DS signal from the CSI analysis [36].

In addition, the results of statistical calculations revealed spatially clustered deformation signals that had similarities in terms of geographical, topographical, and other conditions. The insignificant points in Fig. 11(a) were the results of MP selection by the OHSA method; the lack of a spatial or topographical relationship prevented these points from becoming a deformation cluster. In Fig. 11(b), the cluster

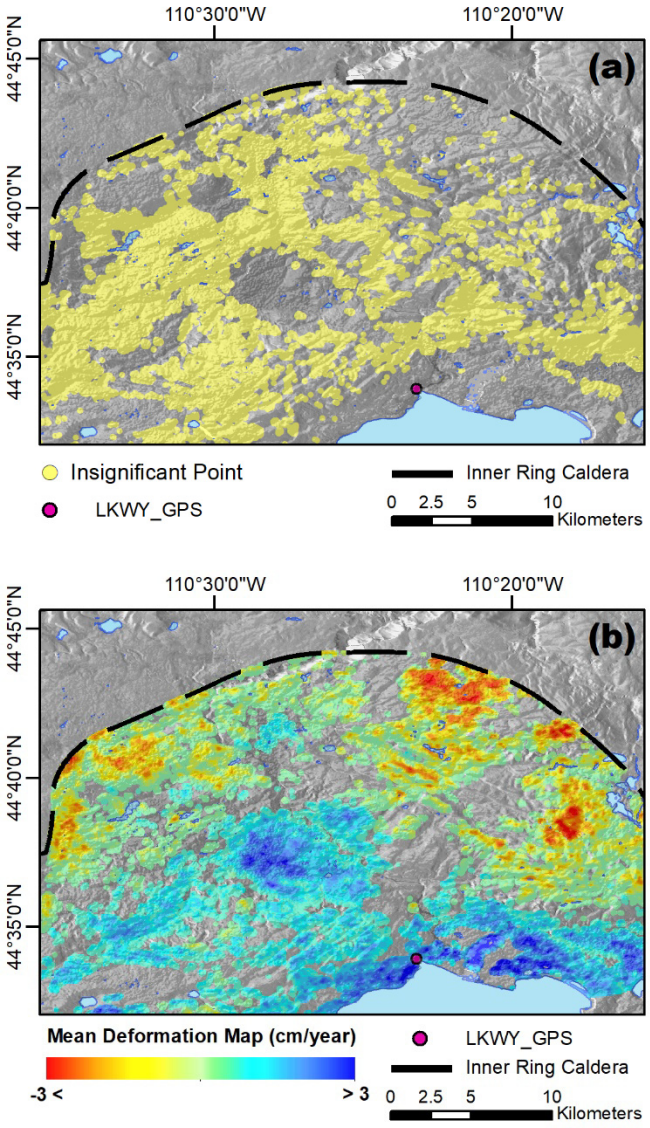


Fig. 11. Mean deformation map of the study area from 2017–2020: (a) insignificant points from optimization method and (b) ICOPS method.

in red indicates a group of MPs experiencing subsidence, which was found in the southwest of the study area around Pelican Valley and Sour Creek. The clusters in blue were MP groups with increasing deformation rates, which were found in the center of the study area (e.g., Hayden Hills); several MP clusters were also located on the shores of Yellowstone Lake. This was advantageous in terms of analyzing deformed areas, especially for low-rate deformations (e.g., landslides or subsidence). It also improved the reliability of detection as the initial step in the risk assessment process. Although there was an increase in the number of MPs detected by the CSI method, the proposed optimization process also improved surface deformation analysis in terms of extracting MP information. The OHSA optimization process separated MPs into several categories, such as insignificant points and clusters of deformation points. This separation was obtained using a 99% confidence value or a p-value $= 0.01$ in the clustering process

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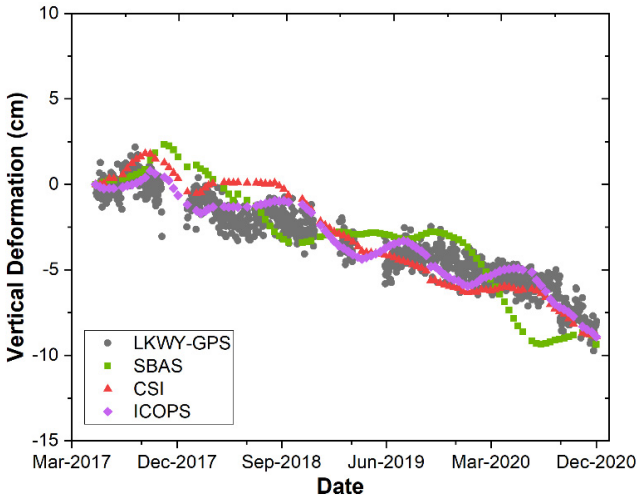


Fig. 12. Time-series graph from LKWY station compared with SBAS, CSI, and ICOPS.

for distinguishing clusters of insignificant points and spatial cluster deformation points.

C. Validation of Deformation Time-Series Using GPS Measurements

It is difficult to quantify the error of an InSAR measurement because phase delays of the troposphere and ionosphere cannot be entirely avoided [49]. In this study, low- and high-pass filterings were applied in the proposed time-series analysis method to reduce atmospheric disturbance. The displacement measurement of the time-series method was validated with the displacement data from GPS at the LKWY station provided by UNAVCO [50]. Based on a time-series analysis using the SBAS and CSI methods, we extracted the deformation time series during the observation year, as shown in Fig. 12. Unfortunately, we could not obtain the displacement time series at the LKWY station for use as ground truth data in the StaMPS analysis because detected PS were lacking in that area. Based on the CSI and SBAS results, the mean deformation rates at the LKWY station were 2.34 and 2.78 cm/year, respectively. The ICOPS result indicated that the mean deformation rate at the LKWY station was 2.23 cm/year, with slight fluctuations in the time-series graph in Fig. 12, which were stacked with the CSI method. Based on the GPS data, the surface deformation mainly occurred near the LKWY station, with fluctuations apparent on the time-series graph. The seasonal deformation that occurred in this area may be an indicator of the geologic activity or climate conditions during the observation period. The possible causes of deformation are considered in Section V.

The statistical properties are provided in Table I. The results of time-series analysis revealed discrepancies in the displacement rate between the InSAR and GPS determinations of <1 cm/year. The standard deviations for CSI and SBAS were 0.945 and 0.99 cm/year, respectively. Based on the statistical features, the standard deviations for CSI and ICOPS were 0.94 and 0.76 cm, respectively. Therefore, the new method based on the CSI process achieved a 13% decrease in the standard deviation.

TABLE I
COMPARISON OF STATISTICAL CALCULATIONS FOR SBAS, CSI, AND ICOPS

	SBAS (cm/year)	CSI (cm/year)	ICOPS (cm/year)
Mean velocity	**2.78 ± 0.99**	**2.34 ± 0.945**	**2.23 ± 0.76**
RMSE	1.42	1.2	1.08

We also calculated the root mean square error (RMSE) of the two methods from GPS station data, with respective values of 1.42 and 1.2 cm/year obtained for SBAS and CSI. The RMSE calculation indicated that the CSI had an initial advantage in the deformation time-series analysis. In the RMSE calculation for measurement of the ICOPS time-series method with GPS, we obtained a value of 1.08 cm/year for the ICOPS method, which showed that the RMSE value decreased by 12.8%, compared with the CSI result; it was 23% smaller than the SBAS result. The smaller RMSE value indicated an improved level of accuracy for the new method.

The differences between InSAR and GPS measurements may be caused by several factors. One such factor was the spatial mismatch between the InSAR MP and the GPS station. As shown in Fig. 12, there was variance in the surface displacement found at the LKWY station, with significant changes in measured displacement values over a distance of several meters. In summary, there was a good overall agreement between the different InSAR methods and the InSAR and GPS measurements, which showed that the proposed new CSI method could achieve accurate and reliable surface displacement measurements. Compared with GPS observations, the InSAR technique provided a higher density of MPs, equivalent to thousands of GPS stations per km^2 [29]. However, the CSI method detected more MPs than did the PS in rural areas.

To evaluate consistency among results obtained by SBAS, CSI, and ICOPS, pairwise correlation analyses were conducted using deformation rate measurement values generated by the three methods using Sentinel-1 data. Fig. 13 shows the correlation coefficients in scatter plot form for each algorithm. For each pair, only the ordinary MPs were plotted and used to calculate the Pearson correlation coefficient. Several points can be inferred from Fig. 13. First, for the stacked measurement data, the correlation between CSI and ICOPS ($R^2 = 0.965$) was higher than the correlation between CSI and SBAS ($R^2 = 0.915$). This was consistent with the significant differences in MP detection capabilities among the three methods. Second, the correlation analysis results indicated that the time series analysis processes could be easily compared; only slight differences were evident.

V. DISCUSSION

A. CSI InSAR Time-Series Analysis

When an InSAR time-series technique is used for the detection of deformation on steep and woodland terrain, the low spatial density of detectable coherence points usually

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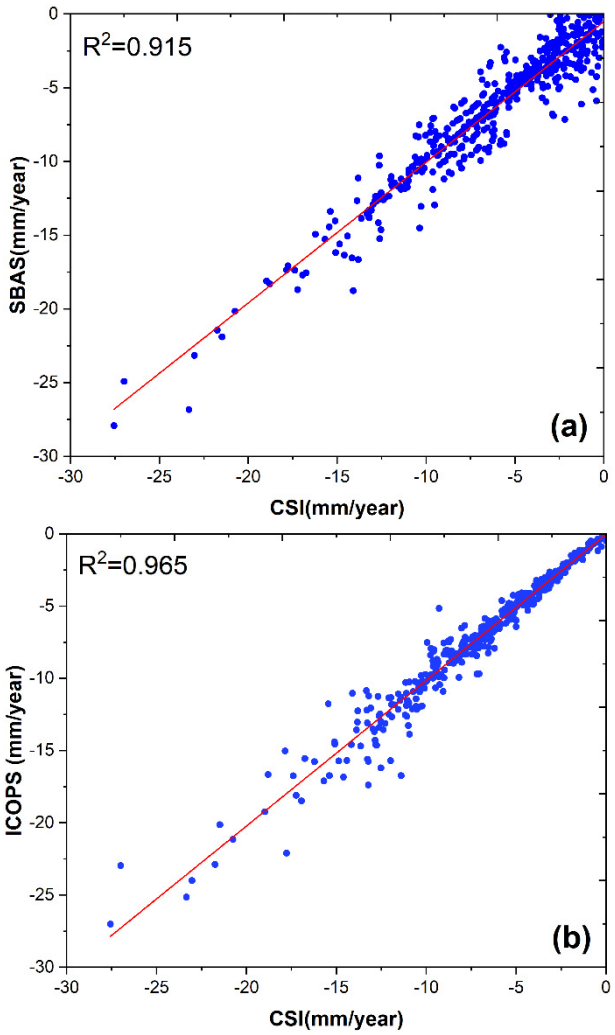


Fig. 13. Correlation coefficients from mean deformation map for relationships between (a) CSI and SBAS and (b) CSI and ICOPS.

leads to phase unwrapping errors; this produces substantial uncertainty in deformation measurements [12]. The proposed ICOPS based on the CSI method combined PS and DS to increase the number of MPs; it also implemented a unique preprocessing strategy for DS target identification and optimal phase estimation. The significant increase in MP density effectively improved the measurement of deformation by reducing the phase unwrapping error and increasing the measurable spatial gradient of the deformation. Using CSI, our ability to discover potential surface deformation was greatly enhanced because CSI could also identify small unstable phases that were difficult to detect in a traditional InSAR time-series analysis. Furthermore, the adaptive filtering approach based on SHP calculation had an influence in terms of the unwrapping phase process. The traditional approach mostly uses rectangular averaging based on window size to filter the interferograms [29], [34].

However, regardless of CSI method robustness, DS target preprocessing requires considerably more time than other traditional time-series methods [27]. In our study, *C*-band wavelength Sentinel-1 SAR data were processed for analysis. Compared with X-band and L-band data which provide greater resolution, it may influence the data processing and create challenges in CSI analysis. Nevertheless, the large amount of information associated with the rapid creation and accumulation of SAR data archives will require a more efficient InSAR data processing system. Thus, there is an urgent need for the development of massively parallelized and integrated CSI methods to meet InSAR time-series data analysis requirements.

B. ICOPS Measurement Process

After the implementation of the optimization process with respect to MPs obtained in the ICOPS process using SAR Sentinel-1 data, clear time-series analysis performance improvements were achieved. These results demonstrated the usefulness of the optimization process using machine learning on MPs to detect surface deformation changes. There were two general advantages of this optimization process: optimization of MPs with machine learning and spatial clustering in the deformation detection process. The use of the SVR algorithm to identify the optimal MP was advantageous when searching for a more flexible and suitable solution for nonlinear datasets. The regression method selected the optimal MP with the SVR algorithm by minimizing errors when searching for the optimal hyperplane with the shortest distance from all datasets [51]. This provided flexibility in the solution disbursement process; in addition, the level of accuracy was influenced by parameter values used in the learning process [52]. In this study, the deformation assumption for the inventory dataset treatment in the SVR process was based on a linear process; this assumed that the deformation occurred in linear terms. Therefore, the process needed to develop a deformation assumption for inventory dataset treatments, such as the treatment based on GPS data, to gain further advantages. Although the SVR algorithm showed promise in processing a large number of MP datasets, it remained deficient in terms of data processing time. However, if the area coverage was expanded and the MP datasets were increased, a more efficient MP analysis process would be needed. Deep learning algorithms (e.g., a convolutional neural network with convolution features) and a pooling process for the dataset could be useful to achieve this optimization [53]. The absence of data mining features in the SVR algorithm could increase analysis precision in the dataset.

The spatial clustering performed in this optimization process offers another advantage in the deformation detection process. The spatial clustering enables the description of the number of occurrences in a geographic space, with a high or low density [54]. The use of the OHSA method, which uses the Getis-Ord* statistical analysis approach, provides a natural clustering process that is based on the characteristics of the deformation rate in each MP. The use of an InSAR time-series analysis based on point scatterers enables the identification of homogeneous areas or infrastructure containing areas with uniform patterns in terms of topography, geomorphology, and land cover [2]. In addition, with a high coherence value, groups of MPs can be used to identify structural objects or

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landscapes that may exhibit a uniform pattern; such patterns will be represented in spatial clusters [16]. Therefore, in areas with MPs that have uniform movements and high coherence, spatial clustering will occur. These results also indicate that OHSA can statistically recognize the spatial clustering of MPs at a high rate for the purpose of early surface change detection.

C. Possible Deformation Characteristics

The northwestern part of Yellowstone Lake subsided throughout the observation period. Recent studies have shown that the continuous subsidence of Yellowstone can be explained by several mechanisms, such as hydrothermal activity and the migration of fluids underground (e.g., water and gases). Large caldera systems, such as Yellowstone, can engage in considerable and far-reaching volcanic activity although events with the greatest possible impact are often rarest [55]. Various modes of minor volcanic unrests are more common, such as earthquake swarms, ground deformation, and hydrothermal activity [56]. Swarms of earthquakes, ground deformation, and changes in the thermal area that would trigger an alarm in dormant smaller stratovolcanoes elsewhere are part of the regular background activity in Yellowstone. Based on the research perspective [57], deformation in the Yellowstone caldera area can be associated with magma intrusion processes in deep rock to volatile releases, propagation, and accumulation in shallow reservoirs that affect the surface above. In addition to hydrothermal activity and fluid flow related to magmatic volatility in the steamboat area around the Norris Geyser Basin (NGB), it is common that it can affect the surrounding deformation. The period of subsidence and magma volatility around the Sour Creek resurgent dome began in March 2014 west of the NGB. However, there are areas that show seasonal deformation signals around the caldera (such as on the shores of Lake Yellowstone), which are associated with the deposition of shallow fault sections with permeable characteristics during the melting of the accumulated snow cover. In this case, seasonal deformation in the southern part of the Sour Creek resurgent dome is in line with periodic reservoir recharge and withdrawal activities. Subsidence occurs during the summer months (May–August) because of lake loading; uplift occurs in late September [22], [58]. This seasonal variation is observed during most of the summer and may be related to groundwater recharge or other environmental factors. The changes were insufficient to influence the overall caldera deformation pattern observed since 2015 [21].

However, the hazards present in active large geothermal caldera systems, such as Yellowstone, should not be overlooked. Hydrothermal explosions occur in the geothermal caldera system; they can be in the form of nonvolcanic ejections of steam, water, and rock [19]. Such explosions occur where pockets of hot water under the surface suddenly turn to steam; this steam breaks up the overlying rock, expels debris, and forms craters. Although the explosions are infrequent, they can cause changes in terms of surrounding surface deformation, such as basin formation or debris overflow. The last large hydrothermal explosion, which formed the 500-m-diameter Indian Pond adjacent to the north side of Yellowstone Lake, occurred approximately 3 000 years ago. Small hydrothermal explosions are estimated to occur at two-year intervals [20].

VI. Conclusion

In this study, a new approach to the analysis of surface deformation changes with multitemporal InSAR (i.e., ICOPS) was proposed to map changes in surface deformation in rural areas and other areas with vegetation cover. The ICOPS was based on the CSI method, which combines and uses PS and DS targets to build a dense observation network for a more reliable unwrapping phase. The GLRT method was used in DS preprocessing as an alternative to the Kolmogorov–Smirnov test, with the aim of obtaining sufficient SHP pixels from minimal SAR data. For PS preprocessing, the StaMPS method was applied to a single-reference interferogram dataset from 2017 to 2020. The minimum cost flow algorithm was then used to estimate the unwrapping phase in the CSI method by estimating the deformation time series using singular value decomposition. However, because MP density increases in a study area after the use of the CSI method, a method to reliably obtain useful information is required. Therefore, a machine learning-based time-series optimization method was proposed in this study. The MPs obtained through CSI were evaluated using the SVR algorithm to obtain the optimal points based on the deformation model obtained. To increase the reliability of deformation detection, a statistics-based OHSA approach was applied to obtain spatial clusters of MPs. This approach enabled deformation detection with spatial clustering, thus allowing the detection of deformation areas even at a low rate.

The CSI method also had advantages in terms of MP detection in vegetation and mountainous areas; this was related to the superiority of using DS points in the analysis. The mean deformation rate at the LKWY station was 2–3 cm/year for the SBAS and CSI methods, while the PSI method did not provide consistent points for analysis. The limitations that arise with the increasing number and density of MPs detected by the CSI method could be overcome by the proposed ICOPS method. The optimized results showed a decrease in the standard deviation of approximately 13% and an approximately 12% decrease in the RMSE value compared with the GPS measurement of the value without optimization. MP spatial clustering was an additional beneficial outcome such that deformation clusters were identified based on the statistical characteristics of the points. There was no need for human intervention when determining the operational parameters. This confirms the potential application of ICOPS as a reliable tool to complement traditional geodetic methods for detecting surface deformation. Although the proposed new approach and its optimized form have strong potential for detecting changes in surface deformation, some limitations were apparent, such as the time-consuming preprocessing stage and the SVR algorithm; these limitations do not permit an independent data extraction function; and they determine the parameters that can affect the results. The deep learning algorithm approach in the optimization process can be used to overcome these problems. However, the ICOPS process proposed in this study is an improvement in the optimization process and will be the focus of continuing development efforts.

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